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2025 Scheme

BAMCS201

RNS INSTITUTE OF TECHNOLOGY

Autonomous Institution Affiliated to VTU
RR Nagar, Bengaluru-560098

II Semester B.E. Degree Examination – June / July 2026
Numerical Methods (CSE stream)

Time: 3 hrs **Max. Marks: 100**

Note:
1. *Answer any 5 full questions, selecting at least one question from each module.*
2. *Missing data may be suitably assumed.*
3. *M: Marks, L: Bloom's level, C: Course Outcomes*

Q. No.		Module-1	M	L	C
Q1	a	Use Secant method to find the root of the equation $xe^x - 2 = 0$ that lies between 0.5 and 1. Perform 3 iterations.	6	L3	CO1
	b	Find real root of the equation $x^3 - 2x - 5 = 0$ that lies in $(2, 3)$ using Regula-Falsi method. Perform 3 iterations.	7	L3	CO1
	c	The true value of $\sqrt{2}$ is 1.414213. An approximation is 1.414. Find the error, absolute error, relative error and percentage error.	7	L2	CO1
		OR			
Q2	a	Find the root of the equation $2x^3 + x - 2 = 0$ using bisection method that lies in $(0, 1)$. Carry out five iterations.	6	L3	CO1
	b	Find the real root of the equation $x \log_{10} x = 1.2$ that lies near 2.5 using Newton-Raphson Method. Carry out 3 iterations.	7	L3	CO1
	c	Derive Regula-Falsi iterative formula to find an approximate root of the equation $f(x) = 0$	7	L2	CO1
		Module-2			
Q3	a	Find the largest eigen value and the corresponding eigen vector of $\begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$ by using Power method. Start with $[1, 0, 0]^T$ as initial approximation.	6	L3	CO2
	b	Use the Gauss –Seidel iterative method to solve the system of equations $5x + 2y + z = 12, x + 4y + 2z = 15, x + 2y + 5z = 20$	7	L3	CO2
	c	Find L_1, L_2 and L_∞ norms of $(2 \quad 3 \quad -4 \quad 6)$	7	L3	CO2
		OR			
Q4	a	Solve the following system of equations by LU decomposition method:	6	L3	CO2

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		$x + 5y + z = 14, 2x + y + 3z = 13, 3x + y + 4z = 17$.			
	b	Find the eigen values and eigen vectors of the symmetric matrix using Jacobi method for the matrix $A = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$.	7	L3	CO2
	c	Find L_1 and L_∞ of $\begin{bmatrix} 2 & -3 & 1 \\ 4 & 0 & -2 \end{bmatrix}$.	7	L3	CO2
		Module-3			
Q5	a	From the following table, estimate the number of students who obtained marks between 40 and 45 by using Newton's interpolation formula Marks | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 No. of students | 31 | 42 | 51 | 35 | 31	6	L3	CO3
	b	Using Newton's divided difference interpolation, find $f(8)$ for the following data: x | 4 | 5 | 7 | 10 | 11 | 13 y | 48 | 100 | 294 | 900 | 1210 | 2028	7	L3	CO3
	c	Obtain the piecewise linear interpolating polynomial from the following table x | 0 | 1 | 2 | 3 y | 1 | 2 | 5 | 10	7	L3	CO3
		OR			
Q6	a	Obtain the piecewise quadratic interpolating polynomial from the following table x | −2 | −1 | 1 | 2 | 4 y | −29 | −8 | −2 | −5 | 7	6	L3	CO3
	b	The following table gives the normal weights of babies during first eight months of life: Age (in months) | 0 | 2 | 5 | 8 Weight (in pounds) | 6 | 10 | 12 | 16 Calculate the weight of baby at age of 7 months using Lagrange's interpolation formula.	7	L3	CO3
	c	Find $f(82)$ using appropriate Newton's interpolation formula, given that $y(40) = 184, y(50) = 204, y(60) = 226, y(70) = 250, y(80) = 276, y(90) = 304$.	7	L3	CO3
		Module-4			
Q7	a	Solve: $\frac{dy}{dx} + \frac{y}{x} = x^2y^6$.	6	L3	CO4

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	b	Solve: $(4D^4 - 8D^3 - 7D^2 + 11D + 6)y = 0$	7	L3	CO4
	c	Solve: $(D^2 - 4D + 4)y = 8(e^{2x} + sin2x + x^2)$	7	L3	CO4
		OR			
Q8	a	Solve: $(8xy - 9y^2)dx + 2(x^2 - 3xy)dy = 0.$	6	L3	CO4
	b	Solve: $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 4\cos^2 x.$	7	L3	CO4
	c	Solve: $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4y = x^2 + 7x + 9.$	7	L3	CO4
		Module-5			
Q9	a	Using Runge-Kutta method to solve $\frac{dy}{dx} = \frac{y-x}{y+x}, y(0) = 1$ at $x = 0.1$ by taking the step length as 0.1	6	L3	CO5
	b	Evaluate $\int_0^6 \frac{1}{1+x^2} dx$ by taking 6 equal parts using Weddle's rule.	7	L3	CO5
	c	Find $y(0.1)$ and $y(0.2)$ using Taylor's Series method considering up to fourth degree term if $y(x)$ satisfies the equation $\frac{dy}{dx} = x^2y - 1; y(0) = 1.$	7	L3	CO5
		OR			
Q10	a	If $\frac{dy}{dx} = xy + y^2$, $y(0) = 1, y(0.1) = 1.1169, y(0.2) = 1.2773,$ $y(0.3) = 1.5040$, find $y(0.4)$ using Milne's predictor-corrector Method.	6	L3	CO5
	b	Evaluate $\int_0^{0.6} e^{-x^2} dx$ by using Simpson's three-eighth rule by taking 7 ordinates.	7	L3	CO5
	c	Using Modified Euler's method find $y(1.2)$, given, $\frac{dy}{dx} = \log_{10}(x + y)$, $y(1) = 2$. Take $h = 0.2$. Carry out two modifications.	7	L3	CO5

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RR Nagar, Bengaluru-560098

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Department of Mathematics

CIE – Test 1

Sem: II
Subject: Numerical Methods (CSE Stream) – for Chemistry cycle
Subject Code: BAMCS201

Date: 15.04.2026
Time: 9:30 am - 11:00 am
Max. Marks: 50

Instruction: Answer any 5 full questions, selecting ONE question from each part.

Q.No.		Question	Marks	RBT*	COs
1	a	Using Newton Raphson iterative formula, find the real root of the equation $x \sin x + \cos x = 0$ that lies near $x = \pi$ correct to four decimal places.	5	L3	CO1
	b	The true value of π is 3.141592. An approximation is 3.142. Find the error, absolute error, relative error and percentage error.	5	L3	
		OR			
2	a	Solve the equation $e^{-x} = 3 \log x$ upto 4 decimal places using secant method lies between 1 and 2. Perform 4 iterations.	5	L3	CO1
	b	Find an approximate root of the equation $x^3 - 3x + 4 = 0$ using the method of false position, correct to four places which lie between -3 and -2. Carry out four iterations.	5	L3	
3	a	Derive the Newton – Raphson iterative formula to find the approximate root for the equation $f(x) = 0$.	5	L2	CO1
	b	Apply Bisection method to obtain the smallest positive root of the equation $x^3 - 5x + 1 = 0$. Perform 3 iterations.	5	L3	
		OR			
4	a	Explain the absolute error, relative error and percentage error and write the corresponding errors if x is rounded off to three decimal places given $x = 0.005998$	5	L2	CO1
	b	Compute the fourth root of 12 correct to three decimal places lies between 1.5 and 2, using Regula – Falsi method. Perform four iterations.	5	L3	
5	a	Given, $\sin 45^\circ = 0.7071$, $\sin 50^\circ = 0.7660$, $\sin 55^\circ = 0.8192$, $\sin 60^\circ = 0.8660$, find $\sin 48^\circ$ using Newton's forward interpolation formula.	5	L3	CO3
	b	Using Newton's divided difference formula, evaluate $f(8)$ from the following:	5	L3	

x	4	5	7	10	11	13
$f(x)$	48	100	294	900	1210	2028

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OR

6 a The area A of a circle of diameter d given for the following values

d (m)	80	85	90	95	100
A (sq. m)	5026	5674	6362	7088	7854

Calculate the area of a circle with diameter 105 m. — 5 — L3 — CO3

b Obtain the piecewise interpolating linear polynomial for the function $f(x)$ defined by the following data:

x	0	1	2	3
y	1	2	5	10

5 — L3

7 a Evaluate $f(7)$ for the data using Newton's divided difference formula:

x	2	4	5	6	8	10
y	10	96	196	350	868	1746

5 — L3 — CO3

b Obtain the piecewise interpolating quadratic polynomial for the function $f(x)$ defined by the following data

x	-2	-1	1	2	4
y	-29	-8	-2	-5	7

Hence find $f(0.5)$ and $f(3)$ — 5 — L3

OR

8 a Find $y\ at\ x = 5$ if $y(1) = -3, y(3) = 9, y(4) = 30, y(6) = 132$ using Lagrange's interpolation formula. — 5 — L3 — CO3

b Obtain the piecewise interpolating linear polynomial for the function $f(x)$ defined by the following data

x	-2	0	1	3	4
y	-23	1	4	82	193

Hence find $f(-0.5)$ and $f(2)$. — 5 — L3

9 a Solve the system of equation by LU decomposition method

$$x + y + z = 1,\ 4x + 3y - z = 6,\ 3x + 5y + 3z = 4$$

10 — L3 — CO2

OR

10 a Apply the Gauss-Seidel iterative method to solve the system of equations

$$2x + 10y + z = 13$$
$$2x + 2y + 10z = 14$$
$$10x + y + z = 12$$

Perform 4 iterations by writing the solutions upto 4 decimal places. — 10 — L3 — CO2

* Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create

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Estd : 2001

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Department of Mathematics

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CIE – Test 2

Sem: II
Course title: Numerical Methods (CS and allied branches)
Course Code: BAMCS201

Date: 29/05/2026
Time: 9:30-11:00
Max. Marks: 50

Instruction: Answer any 5 full questions, selecting ONE question from each part.

Q.No.		Question	Marks	RBT*	COs
1	a	Define matrix norms. Find L_1 and L_∞ of $A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & -1 \end{bmatrix}$	5	L3	CO2
	b	Find the largest eigen value and the corresponding eigen vector of the matrix using Rayleigh power method: $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ taking $X_0 = (1\ 0\ 0)^T$. Carryout 6 iterations.	5	L3	
		OR			
2	a	Check whether the system of equations $2x + y = 2, 2x + 1.01y = 2.01$ are ill defined.	5	L3	CO2
	b	Find the eigen values and eigen vectors of the symmetric matrix using Jacobi method for the matrix $\begin{pmatrix} 6 & \sqrt{3} \\ \sqrt{3} & 4 \end{pmatrix}$	5	L3	
3	a	Solve by Taylor Series method to find the approximate value of $y(0.2)$ given $\frac{dy}{dx} = 3e^x + 2y, y(0) = 0$ considering up to 5th degree term.	5	L3	CO5
	b	Evaluate $\int_0^{0.3} (1 - 8x^3)^{\frac{1}{2}} dx$ taking 3 equal divisions using Simpson's 3/8th rule.	5	L3	
		OR			
4	a	Solve by modified Euler's method $\frac{dy}{dx} = 3x + \frac{y}{2}$, $y(0) = 1$ to find $y(0.2)$ in steps of 0.2 . Perform 3 modifications.	5	L3	CO5
	b	Evaluate $\int_0^{0.6} e^{-x^2} dx$ by taking 7 ordinates using Trapezoidal rule.	5	L3	
5	a	Solve by Runge Kutta method of fourth order $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$, $y(0) = 1$ to find $y(0.1)$ considering $h = 0.1$.	5	L3	CO5
	b	Evaluate $\int_2^8 \frac{dx}{\log_{10} x}$ by taking 7 ordinates using Simpson's 1/3rd rule	5	L3	
		OR			

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6	a	Solve by Milne's predictor corrector method $\frac{dy}{dx} = 2e^x - y,\ y(0) = 2,$ $y(0.1) = 2.01,\ y(0.2) = 2.04,\ y(0.3) = 2.09$ to find $y(0.4)$.	5	L3	CO5
	b	Evaluate $\int_{0.2}^{1.4}(\sin x - \log x + e^x)\,dx$ using Weddle's rule taking 6 equal divisions.	5	L3	
7	a	Solve $(5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0$	5	L3	CO4
	b	Solve $\cos^2 x \frac{dy}{dx} + y = \tan x$	5	L3	
		OR			
8	a	Solve $\frac{dy}{dx} + xy = xy^3$	5	L3	CO4
	b	Solve $(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$	5	L3	
9	a	Solve $(4D^4 - 8D^3 - 7D^2 + 11D + 6)y = 0$	5	L3	CO
	b	Solve $\frac{d^3y}{dx^3} - 7\frac{dy}{dx} + 6y = 1 - x + x^2$	5	L3	
		OR			
10	a	Solve $\frac{d^2y}{dx^2} - 4y = \cosh(2x - 1) + 3^x$	5	L3	C
	b	Solve $\frac{d^3y}{dx^3} + y = 65\cos(2x + 1)$	5	L3	

*** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create**

********** *ALL THE BEST* **********

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